Chihyeong Lee | Curriculum Vitae

Seoul National University – Seoul, Korea
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Google Scholar Profile

Research Interests

- Biomechanics, Musculoskeletal Simulation, Reinforcement Learning, Machine Learning & Deep Learning, Wearable Robotics, Rehabilitation Engineering, Gait Training

Education

Seoul National University — **Seoul, Korea**
Ph.D. Candidate in Sport Science, GPA: 3.98/4.00 — *Mar 2023 – June 2025*
Advisor: Prof. Jooeun Ahn

Seoul National University — **Seoul, Korea**
Master of Sport Science, GPA: 3.91/4.00 — *Mar 2021 – Feb 2023*
Advisor: Prof. Jooeun Ahn

Seoul National University — **Seoul, Korea**
Bachelor of Sport Science, GPA: 3.79/4.00 — *Mar 2014 – Feb 2021*

Academic Experience

Sports Engineering Lab, Seoul National University — **Seoul, Korea**
Undergraduate Researcher — *Jul 2020 – June 2025*
Advisor: Prof. Jooeun Ahn

Department of Health and Human Performance, University of Houston — **Texas, USA**
Visiting Scholar — *July 2024*

Publications

Peer-reviewed Journal Articles

- Park SH, **Lee C**, Park H, Ahn J, Lee B. "Independent and synergistic effects of visual biofeedback and phase-specific belt deceleration on affected-leg propulsion in post-stroke split-belt treadmill training." *Journal of NeuroEngineering and Rehabilitation*, 2025 (Under review)
- Hong T[1], **Lee C**[1], Chang, S, Choi E, Kim B, Ahn J, Park Y. "Design of A Fully-Soft Lift-Assist Wearable Suit Powered by Flat Inflatable Artificial Muscles." *IEEE Robotics and Automation Letters.* 2025 Feb 13; IEEE. (IF: 4.6) ([1]Contributed equally) **[Read Full Paper]**
- **Lee C**, Ahn J, Lee B. "The effects of perturbation intensities on backward slip-falls induced by a split-belt treadmill." *Scientific Reports.* 2025 Feb 11; 15(1):5108. (IF: 3.8) **[Read Full Paper]**
- **Lee C**, Ahn J, Lee B. "A systematic review of the long-term effects of using smartphone-and tablet-based rehabilitation technology for balance and gait training and exercise programs." *Bioengineering.* 2023 Sep 28; 10(10):1142. (IF: 3.8) **[Read Full Paper]**

Peer-reviewed Conference Papers

- **Lee C**, Kim H, Kim C, Ahn J, Kim K, Kwon Y. "Evaluation of Kinematic Similarity between Non-Assisted and Assisted Walking with a Hip Exoskeleton Using a Vector Coding Technique ." *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 14–17, 2025.
- **Lee C**, Ahn J, Lee B. "Evaluating the Impact of Different Slipping Velocities on Falls and Non-Falls Induced by a Programmable Split-Belt Treadmill." *47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 14–17, 2025.
- Yoo D, **Lee C**, Ahn J, Lee B. "Age-related adaptation of the body's kinematic responses to unpredictable trip perturbations induced by a split-belt treadmill." *45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, July 24–27, 2023. **[Read Paper]**

Conference Abstracts

- **Lee C**, Ahn J, Lee B. "Referent-Configuration Imitation Learning (RCIL) for Predictive Musculoskeletal Simulation: Faster and more stable gait imitation by controlling referent configurations instead of muscle activations." *10th World Congress of Biomechanics (WCB)*, July 11–15, 2026.
- Rivera A, Park H, **Lee C**, Ahn J, Park SH, Lee B. "Enhancing paretic propulsion post-stroke via real-time visual biofeedback and adaptive dual-belt treadmill control." *55th Annual Meeting of the Society for Neuroscience (SfN)*, November 15–19, 2025.
- **Lee C**, Choi E, Kim M, Ahn J. "Downhill Walking Classification for Gait-Assive Wearable Robot: A Machine Learning and Deep Learning Approach." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Ahn J, Lee B. "Effects of slip velocity and duration on fall incidence: Evidence from unexpected slip perturbations induced by a split-belt treadmill." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Kim H, Kim C, Ahn J, Kim K, Kwon Y. "KINEMATIC COUPLING RELATIONSHIPS BETWEEN HIP AND KNEE JOINTS WHEN WALKING WITH A HIP EXOSKELETON." *49th Annual Meeting of the American Society of Biomechanics (ASB)*, August 13–16, 2025.
- **Lee C**, Choi E, Choi Y, Kim C, Lee J, Min H, Ahn J. "Gait-Assistive Wearable Robots: Biomechanical Evaluation Schema." *2025 Soft Robot Research Center (SRRC) Symposium*, Fev 21, 2025.
- **Lee C**, Choi E, Min H, Ahn J. "Quantitative Evaluation of Familiarization with Walking Assistive Wearable Robots - A preliminary study on hip joint assistive robots." *2024 Korea Society of Sport Biomechanics (KSB)*, Nov 21, 2024.
- **Lee C**, Ahn J. "REAL-TIME BACKWARD SLIP DETECTION USING A SLIP-INDUCING SYSTEM AND MACHINE LEARNING METHODS." *41st International Society of Biomechanics in Sports (ISBS)*, July 12–16, 2023. **[Read Abstract]**
- Yoo D, **Lee C**, Ahn J, Lee B. "ASSOCIATION BETWEEN AGE AND BODYS KINEMATIC RESPONSES TO UNPREDICTABLE GAIT PERTURBATION." *41st International Society of*

Biomechanics in Sports (ISBS), July 12–16, 2023. **[Read Abstract]**

Invited Lecture

- **Lee C**. "Advancing Laboratory Fall Reseach - Toward Fall Prediction and Fall-Prevention Training." Guest Lecture in *Biomechanics of Lower Extremities* course, Seoul National University, Seoul, Korea, Nov 2024.

Projects

- **Robot Industry Core Technology Development Project**
  - *Mar 2021 – Dec 2023*
  - *Aim: Development of fabric-type hybrid actuator with embedded soft sensor and wearable robot technology to support strength of industrial workers*
  - *Role: Robot Development Evaluator*

    1) Developing lightweight and flexible wearable robots to assist lifting tasks in industrial settings through various structural designs and support mechanisms.

    2) Collaborating with a multidisciplinary team to integrate biomechanics and robotics for improving efficiency and safety in industrial work environments.

- **Robot Industry Core Technology Development Project**
  - *Mar 2022 – Dec 2024*
  - *Aim: Development of Wearable Robot Product and Service Platform for Healthcare in Home and Daily-life Environment*
  - *Role: Robot Development Evaluator*

    1) Evaluating wearable robots for lift-assist and walking support in home and daily living environments, and providing feedback for development improvement.

    2) Collaborating with a multidisciplinary team to integrate biomechanics and robotics for enhancing efficiency and safety in home and daily living environments.

- **Brain Pool Project**
  - *Dec 2022 – Nov 2023*
  - *Aim: Detection and prevention of fall using an advanced fall inducing system*
  - *Role: TBD*

    1) Evaluating wearable robots for lift-assist and walking support in home and daily living environments, and providing feedback for development improvement.

    2) Collaborating with a multidisciplinary team to integrate biomechanics and robotics for enhancing efficiency and safety in home and daily living environments.

- **User-Centered Care Robot and Service Demonstration R&D Project**
  - *Mar 2023 – Dec 2027*
  - *Aim: Development of long-term and short-term clinical/biomechanical evaluation indicators and analysis of body effect on flexible wearable care robots*
  - *Role: Robot Development Evaluator*

    1) Evaluating walking assistive wearable robots and developing standardized assessment protocols.

2) Collaborating with a multidisciplinary team to integrate biomechanics, robotics, and clinical assessment for the development of reliable and valid evaluation standards.

- **Brain Pool Project**
  - *Mar 2024 – Dec 2026*
  - *Aim: Development and Clinical Assessment of a Patient-Tailored Intelligent System for Gait Rehabilitation in Stroke Patients via Real-Time Multiple Assistance with Propulsion Visual Feedback and Propulsion Promotion*
  - *Role: TBD*

  1) Evaluating wearable robots for lift-assist and walking support in home and daily living environments, and providing feedback for development improvement.

  2) Collaborating with a multidisciplinary team to integrate biomechanics and robotics for enhancing efficiency and safety in home and daily living environments.

Other Experience

Seoul National University, 2nd Division Team | **Seoul, Korea**
Varsity Basketball Coach | *Sep 2020 – 2022, 2025*
- 3rd Place, National University Basketball Tournament | *2022*

Varsity Basketball Player | *May 2014 – Aug 2019*
- 3rd Place, National University Basketball Tournament | *2019*
- 3rd Place, National University Basketball Tournament | *2017*
- 3rd Place, National University Basketball League | *2014*
- 3rd Place, National University Basketball Tournament | *2014*

Jeju National Hospital | **Jeju, Korea**
- *Social Service Personnel (Mandatory Military Service)* | *May 2018 – Aug 2019*

Gangneung Public Health Center | **Gangneung, Korea**
- *Social Service Personnel (Mandatory Military Service)* | *Sep 2017 – Apr 2018*

Seoul National University Middle School | **Seoul, Korea**
- *Student Teacher* | *May 2017 – Jun 2017*

References

Prof. Jooeun Ahn

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Prof. Beom-Chan Lee

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